ABDULLAH MANZOOR MALLICK

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with a strong systems foundation — final-year CS student who has shipped a Java internship at AKSA Solutions Development Services and built production-grade systems spanning distributed architecture, blockchain, and applied ML. Comfortable owning a problem end-to-end: designed and deployed a microservices banking platform with a Jenkins CI/CD pipeline on Azure, engineered a Lambda-architecture analytics pipeline processing 7M+ records in real time, and built a Zero-Knowledge Proof Ethereum wallet from cryptographic circuit to production UI. Seeking Software Engineering, Backend, or AI/ML roles where deep technical range and self-driven execution are valued.

TECHNICAL SKILLS

Languages: C · C++ · C# · Java · Python · JavaScript · x86 Assembly · SQL · Solidity

Backend & AI/ML: Spring Boot · Spring Security (custom JWT/RBAC) · Spring WebFlux/R2DBC · LangChain · RAG pipelines · ChromaDB · scikit-learn (MLP · RF · DT · KMeans) · PyMC · NumPy · Pandas · PySpark MLlib

Big Data & Messaging: Hadoop (MapReduce · HDFS) · Spark (PySpark · Spark SQL · MLlib) · Kafka · Spring Cloud Stream · Storm · Docker

Blockchain / Web3: Solidity · Hardhat · Foundry · Circom · Groth16 · Ethers.js · Viem · ERC-20 · Ethereum Sepolia · Polygon Amoy

Web & UI: HTML5 · CSS3 · JavaScript · React · Thymeleaf · Java Swing + JDBC · Streamlit

Databases: MySQL · PostgreSQL · MongoDB · SQL Server · SQLite · R2DBC

DevOps / Cloud: Jenkins CI · Microsoft Azure (App deployment) · Docker · Docker Compose · Git/GitHub

Systems & Compilers: OS concepts · x86 Assembly · BSD Sockets (C) · Lexical analysis (DFA/NFA) · LL(1) · LR(1)/SLR(1) · Flex & Yacc/Bison

Security: AES · DES · RSA · PKI · SQL injection · XSS · CSRF · Penetration testing · CTF

EXPERIENCE

Summer Intern (Java) — AKSA Solutions Development Services (Pvt.) Ltd. Islamabad, Pakistan | June – August 2026

- Owned the full build cycle for 8 full-stack Spring Boot systems in 6 weeks — shipping custom JWT auth with fine-grained RBAC (no Spring Security), a reactive WebFlux/R2DBC service, concurrency-safe booking logic under simulated load, and a Kafka event-streaming pipeline.
- Defended architecture and design trade-offs directly to a senior engineer for every project, translating self-taught concepts — SOLID principles, layered architecture, JWT security flows — into reviewed, working code with zero handholding.

PROJECTS

Sentinel Z – ZK-Gated Account-Abstraction Wallet [FYP] Solidity · Circom · Groth16 · React · Hardhat · Ethers.js | Spring 2026

- Designed a tiered-authorization Ethereum wallet from the cryptographic circuit up: ZK-Proof verification (Circom + Groth16) gates high-value transactions while daily-spend-capped key-sig handles routine spending, enforced entirely on-chain.
- Shipped the full stack solo on architecture — React UI, end-to-end test suite, and ERC-20/Uniswap dApp compatibility — deployed live on Ethereum Sepolia and Polygon Amoy testnets. Team of 3.

Real-Time Crime Analytics & Alert System [Big Data Analytics] PySpark · Kafka · Storm · PostgreSQL · MongoDB · Streamlit · Docker | Spring 2026

- Built a Lambda-architecture pipeline processing a 5.7 GB / 7M+ record public safety dataset: PySpark batch layer clustering 7M+ GPS coordinates via K-Means, paired with a Kafka + Storm speed layer flagging anomalies in sub-second sliding windows.
- Delivered a fully containerised serving layer (PostgreSQL + MongoDB + Streamlit) supporting millisecond historical queries alongside live streaming alerts — architected and deployed independently via Docker Compose.

Helix — Distributed Banking Microservices with CI/CD Java · Spring Boot · Jenkins · Azure · Docker

- Architected a microservices-based banking platform and built a Jenkins CI/CD pipeline automating build, test, and deployment — then validated it end-to-end with a live deployment to Microsoft Azure (later decommissioned after free-tier expiry).

CryptoTracker — Reactive Live Price Streaming Service Spring Boot WebFlux · R2DBC · JPA · MySQL · JWT

- Engineered a fully non-blocking pipeline polling live market data every 30s and streaming price updates to clients over Server-Sent Events, with zero blocking calls on the reactive event loop.

- Solved a real architectural conflict — coexisting R2DBC (high-frequency reactive writes) with JPA (batch archival) in one service — by isolating blocking JPA work on a dedicated scheduler thread.

Nexo — Personal Expense Tracker Spring Boot · PostgreSQL · React · Vite · Tailwind · Chart.js · Railway · Vercel

- Built and deployed a full-stack expense tracker live to production — Spring Boot/PostgreSQL backend on Railway, React/Vite/Tailwind frontend on Vercel — with JWT auth, budgeting, spending analytics, and Swagger-documented APIs.

Cognify — RAG Document Q&A Chatbot FastAPI · LangChain · ChromaDB · Groq (Llama 3) · React

- Built a retrieval-augmented generation pipeline: documents are chunked, embedded, and indexed in ChromaDB, then queried through LangChain to ground Llama 3 (via Groq) responses in the source material rather than the model's raw knowledge.
- Served through a FastAPI backend with a React chat interface.

Additional projects

- Signal (Kafka event-streaming service), Inventra (inventory management with custom RBAC), Weather Aggregator (real-time WebSocket data pipeline), Ticket Booking Concurrency Simulator (race-condition handling three ways), Library Management System (SOLID/design-pattern-driven, 22-test suite), Custom Language Compiler (LL(1)/SLR(1) parser + JSON→XML translator), and an Autonomous Racing AI trained via scikit-learn MLP — full details on GitHub and portfolio.

EDUCATION

FAST University NUCES, Islamabad Fall 2022 – Present

BS Computer Science

Coursework: Data Structures & Algorithms · OOP · OS · Databases · Computer Networks · Compiler Construction · Software Engineering · AI · Big Data Analytics · Statistical Modeling · Information Security · Assembly Language · Numerical Computing · Cloud Computing

CERTIFICATIONS

- November 2025 — AWS Cloud Practitioner Essentials — Amazon Web Services
- August 2024 — Pre-Security Path — TryHackMe (THM-ASMDQCTJKT)
- October 2023 — C++ Programming — ICP Institute
- August 2023 — C++ Certification & OOP Concepts in C++ — Programming Hub / Great Learning

ADDITIONAL

- Leadership: Scrum Master & Product Owner (Software Engineering project); team lead across multiple group projects.
- Version Control: active GitHub (Abdullah-07s); Git branching, pull requests, collaborative workflows.
- Cloud & DevOps: AWS Cloud Practitioner Essentials; Jenkins CI/CD pipeline configuration; Azure deployment experience; containerised deployment via Docker Compose.
- Languages: English (professional fluency) · Urdu (native).